In the above code, the name of the class is Sample. Inside the class, we have assigned the variables **x** and **y** with initial value **10** and **20** respectively. These two variables are called as class variables or member variables of the class. In class instantiation process, we have created an object **S** to access the members of the class. The first two print statements simply print the value of class variable **x** and **y** and the last print statement add the two values and print the result.

10.5 Class Methods

Python class function or Method is very similar to ordinary function with a small difference that, the class method must have the first parameter named as **self**. No need to pass a value for this parameter when we call the method. Python provides its value automatically. Even if a method takes no arguments, it should be defined with the first parameter called **self**. If a method is defined to accept only one parameter it will take it as two arguments ie. self and the defined parameter.

When you access class variable within class, methods must be prefixed by the class name and dot operator.

Note

- The statements defined inside the class must be properly indented.
- Parameters are the variables in the function definition.
- Arguments are the values passed to the function definition.

Example: Program to find total and average marks using class

```
class Student:
        mark1, mark2, mark3 = 45, 91, 71          #class variable

        def process(self):                        #class method
            sum = Student.mark1 + Student.mark2 + Student.mark3
            avg = sum/3
            print("Total Marks = ", sum)
            print("Average Marks = ", avg)
            return

S=Student()
S.process()
```

In the above program, after defining the class, an object S is created. The statement S.process( ), calls the function to get the required output.

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